

CAPSTONE PROJECT • 2026

Bluestock Mutual Fund

Analytics Capstone

End-to-End Mutual Fund Analytics and Business Intelligence Solution

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Technologies: Python · Pandas · SQLite · SQL · Power BI · Plotly

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PROBLEM STATEMENT & OBJECTIVES

PROBLEM STATEMENT

The mutual fund industry generates large volumes of data related to fund performance, investor transactions, SIP trends, portfolio allocations, and market benchmarks. Extracting meaningful insights from this data requires an integrated analytics solution.

PROJECT OBJECTIVES

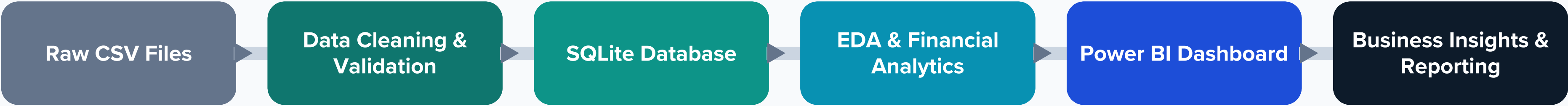
- Clean and validate mutual fund datasets
- Build a centralized analytical database
- Perform exploratory data analysis
- Evaluate fund performance using financial metrics
- Analyze investor behavior and SIP trends
- Create an interactive Power BI dashboard
- Generate actionable investment insights

DATA SOURCES

1	Fund Master	2	NAV History
3	AUM by Fund House	4	Monthly SIP Inflows
5	Category Inflows	6	Industry Folio Count
7	Scheme Performance	8	Investor Transactions
9	Portfolio Holdings	10	Benchmark Indices

◆ 40 Mutual Fund Schemes ◆ 2022–2026 Historical Data ◆ Investor Demographics ◆ Benchmark Market Data

SOLUTION ARCHITECTURE



Tools & Technologies



EDA Highlights – NAV Analysis

Objective: Analyze mutual fund NAV movement over time to identify growth trends and volatility patterns.

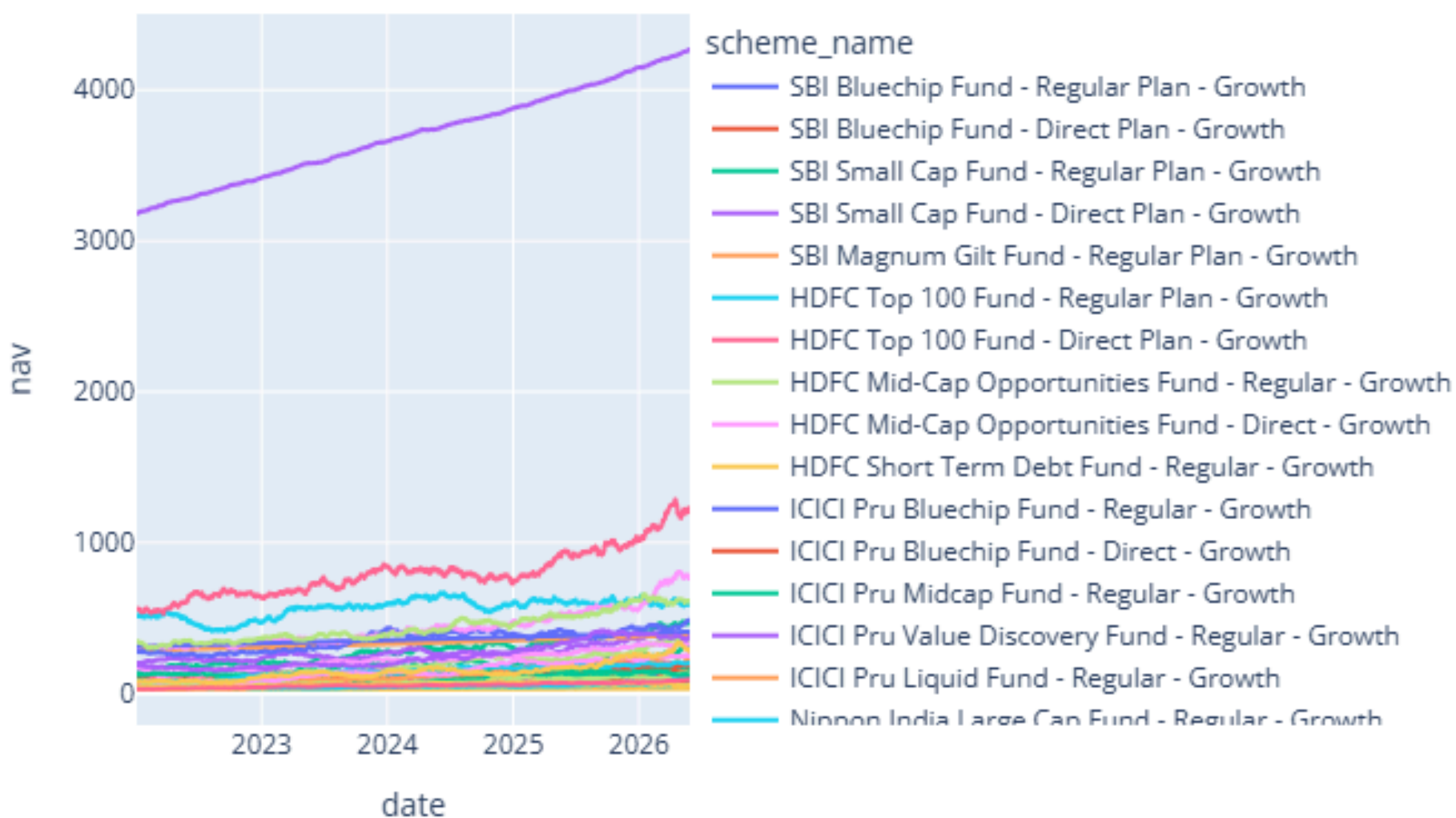
Most equity funds demonstrated consistent long-term growth

Significant market recovery observed after corrections

Large-cap funds exhibited relatively stable NAV trends

Small-cap funds showed higher volatility compared to large-cap peers

Daily NAV Trends



NAV Trend Chart

EDA Highlights – SIP & Industry Growth

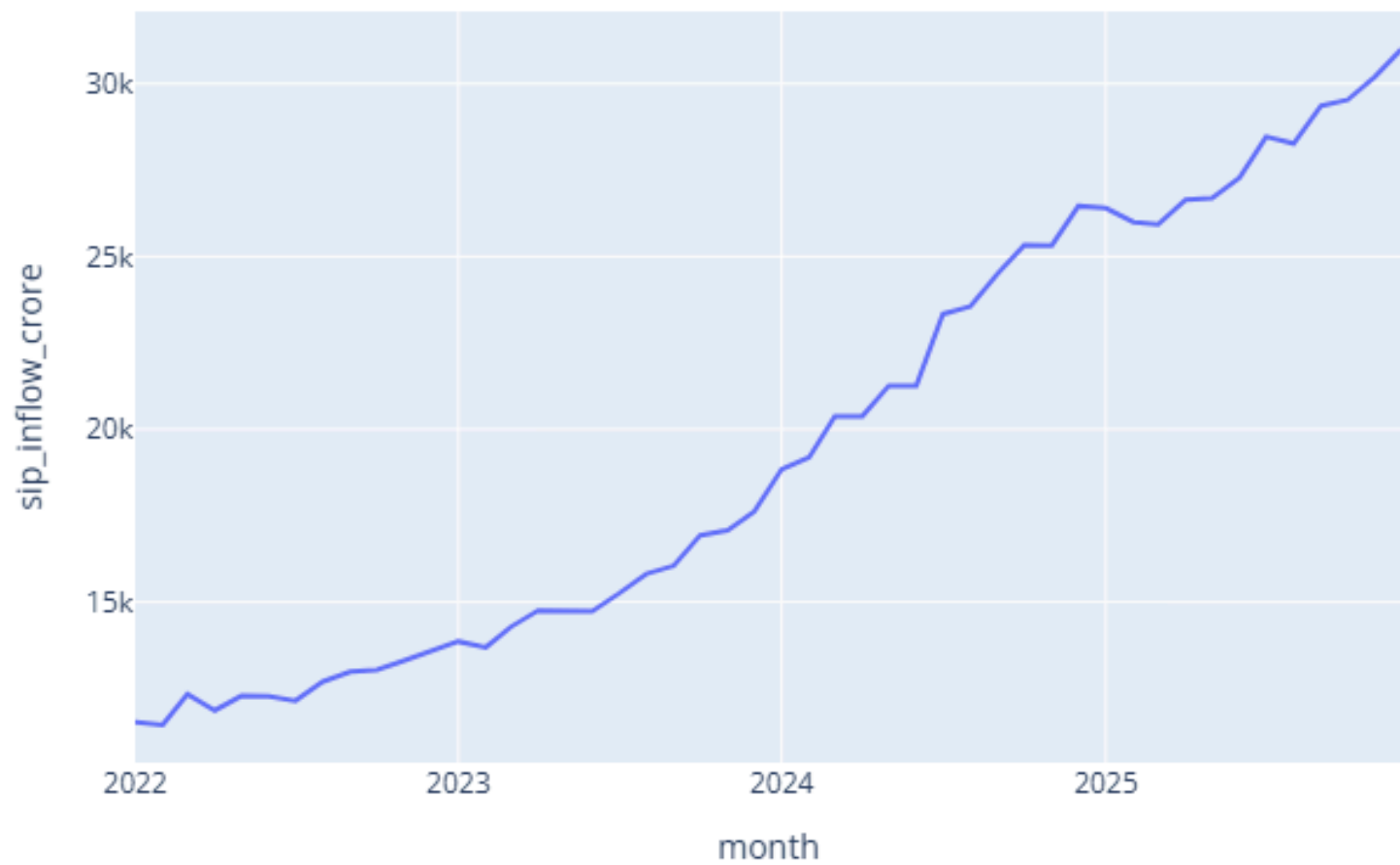
Objective: Study investor participation and SIP growth trends from 2022 to 2026.

◆ SIP inflows increased significantly from 2022 to 2025

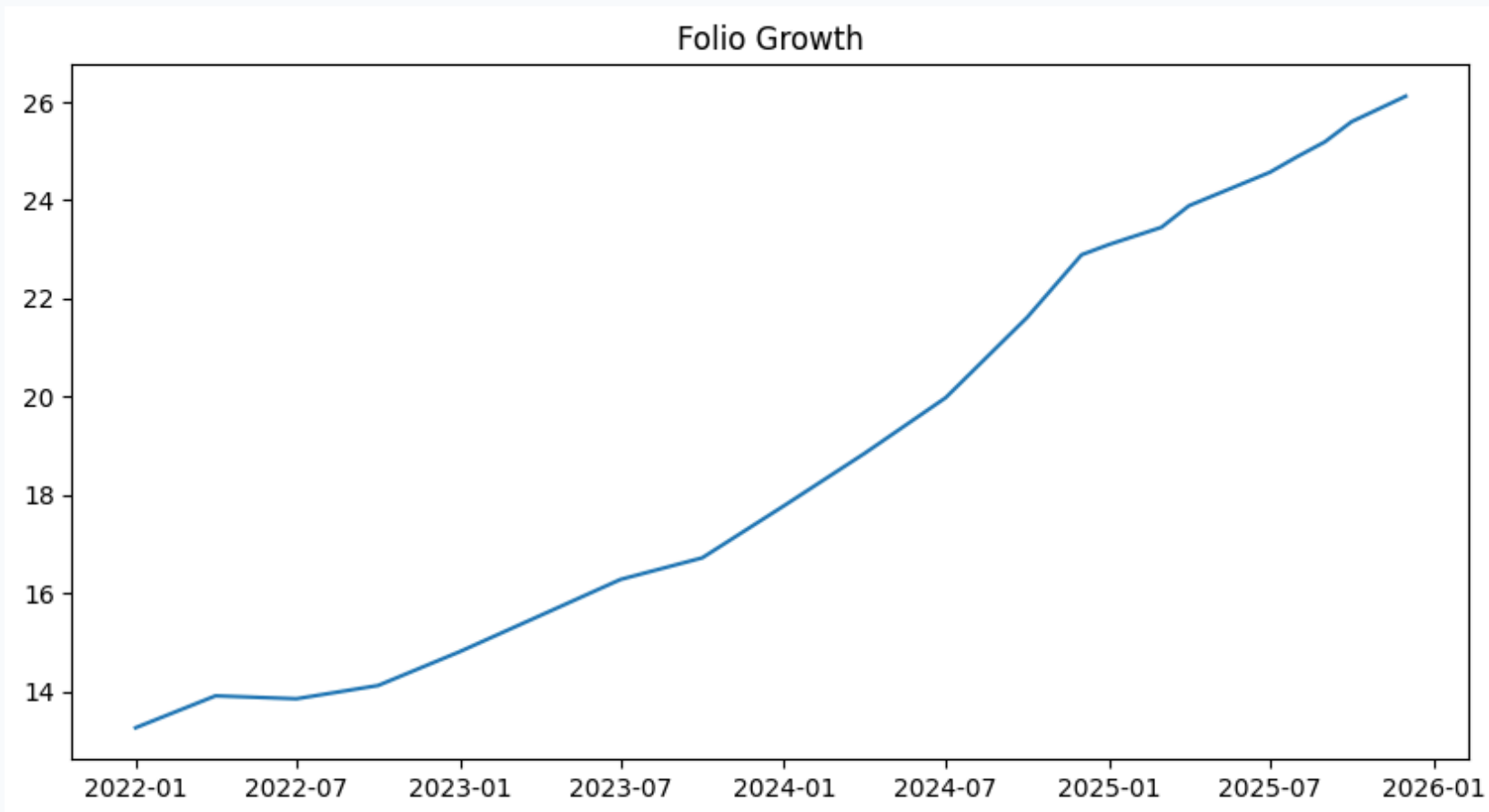
◆ Retail investor participation expanded rapidly

◆ Industry folio count nearly doubled during the analysis period

◆ Mutual fund penetration improved across India



SIP Trend Chart



Folio Growth Chart

Performance Analytics

CAGR

Daily Returns

Sharpe Ratio

Sortino Ratio

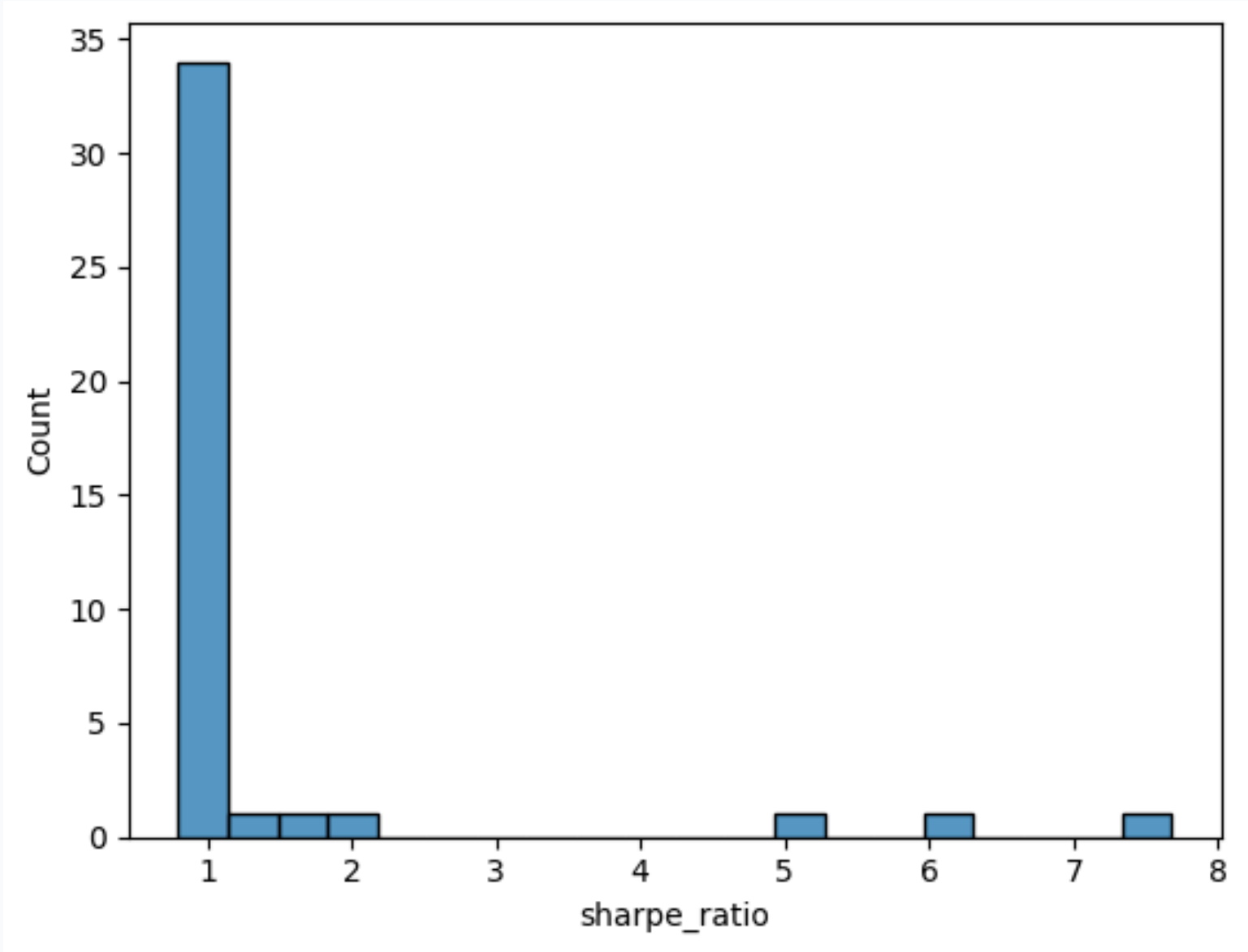
Key Insights

- 1

High Sharpe ratio funds delivered superior risk-adjusted returns
- 2

Consistent performers outperformed peers across multiple periods
- 3

Risk-adjusted ranking provided a better evaluation than returns alone



Sharpe Ratio Ranking Table

Advanced Performance Metrics

Alpha

Beta

Max Drawdown

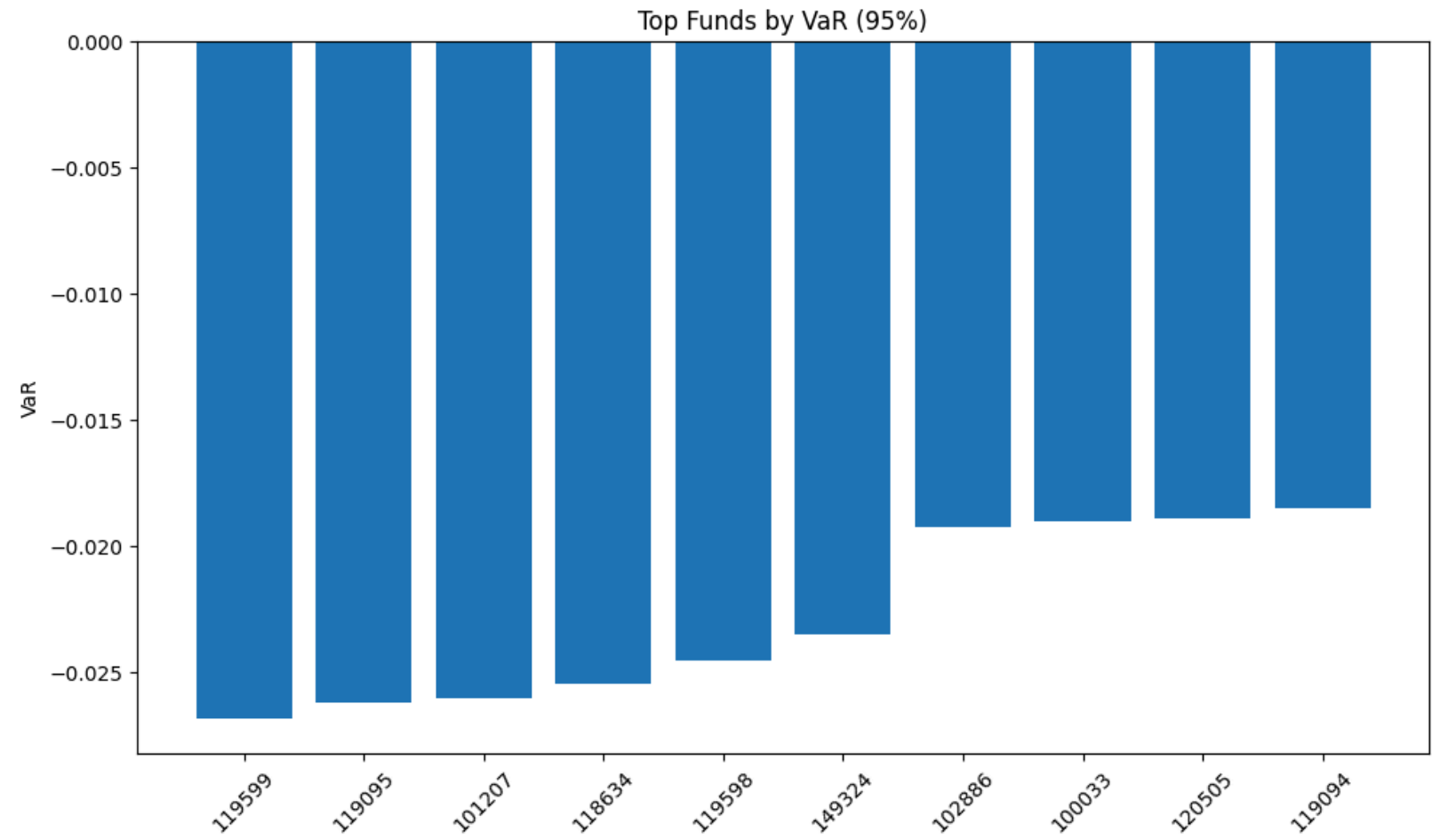
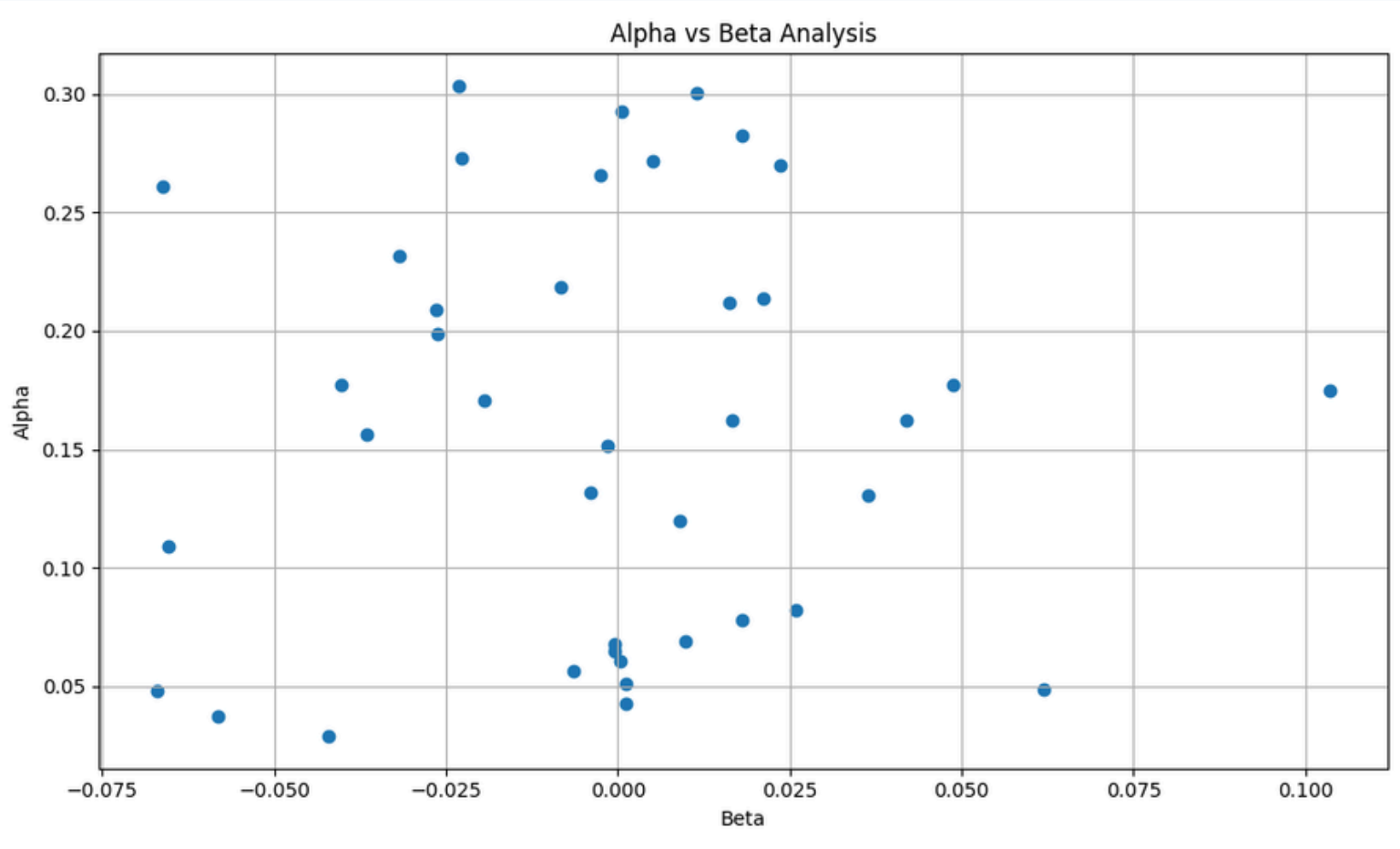
VaR

CVaR

Key Insights

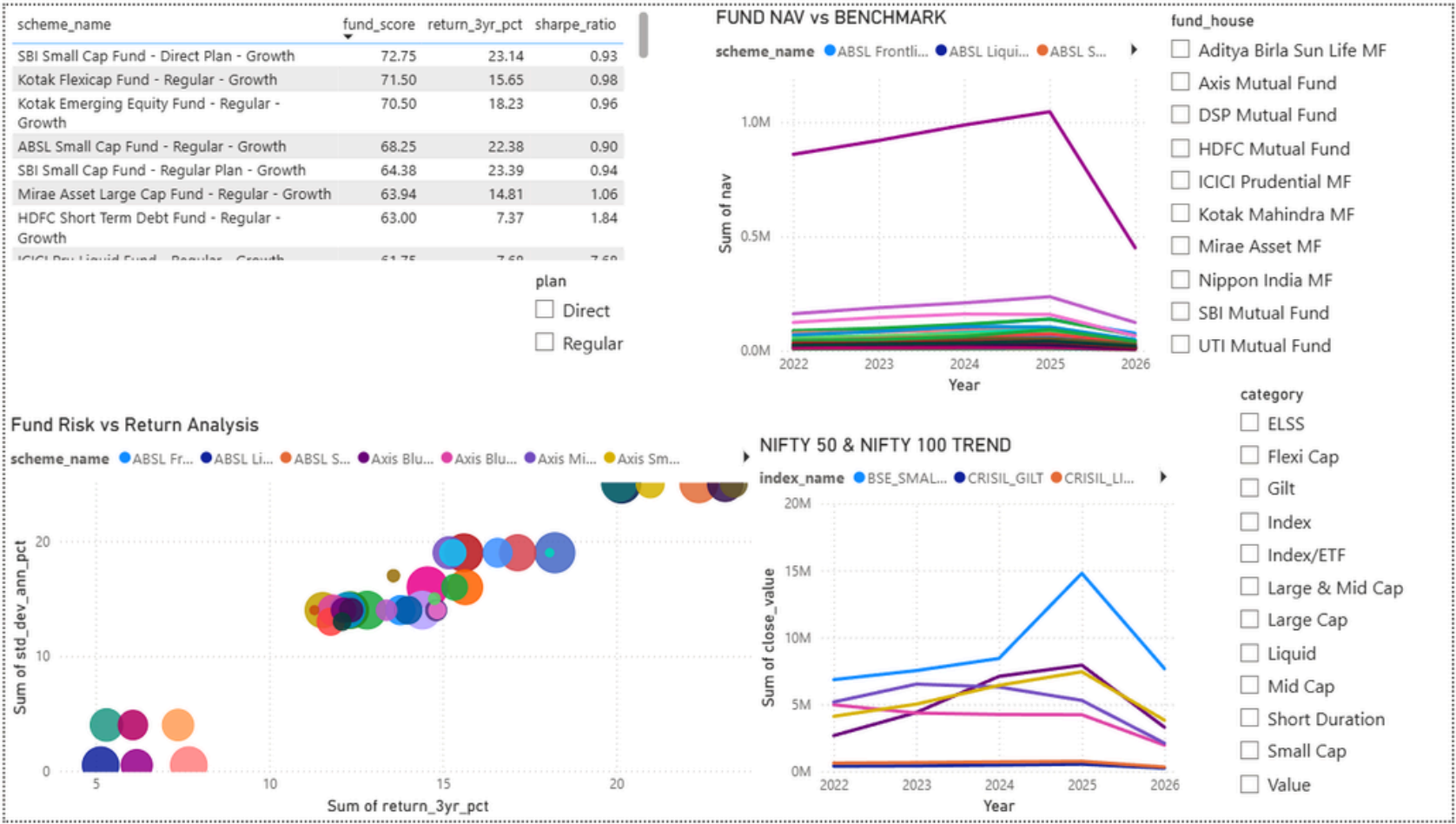
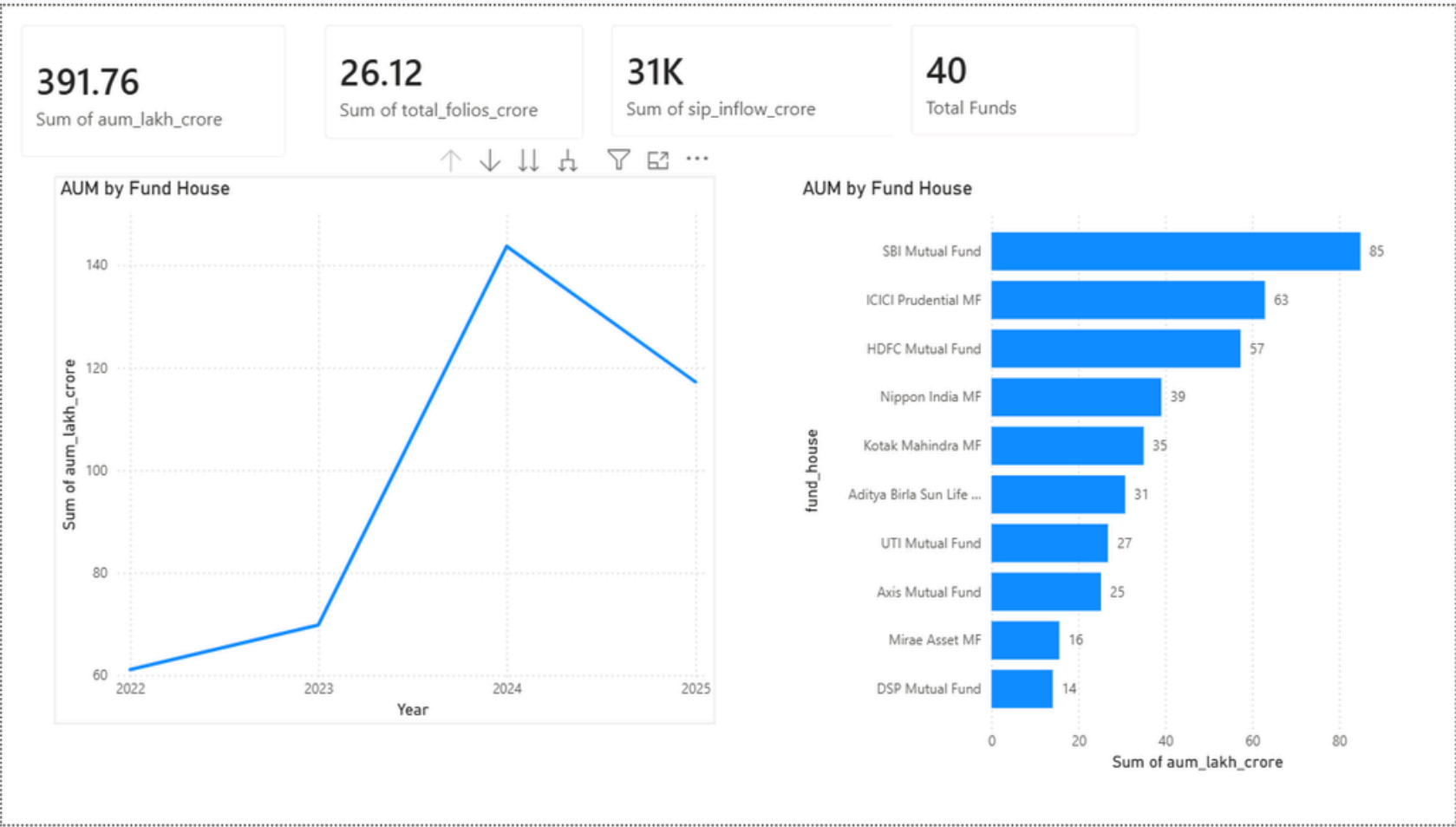
- α Positive Alpha funds consistently outperformed benchmarks
- β Higher Beta funds exhibited stronger market sensitivity

- VaR VaR and CVaR identified downside risk exposure accurately
- ∇ Maximum Drawdown highlighted worst historical losses



Dashboard Overview – Industry & Fund Analytics

Power BI Dashboard – Page 1 & 2



Business Value

Enables quick identification of top-performing funds

Supports fund comparison and risk evaluation

Dashboard Features

◆ Industry KPI Cards

◆ Fund Risk vs Return Visualization

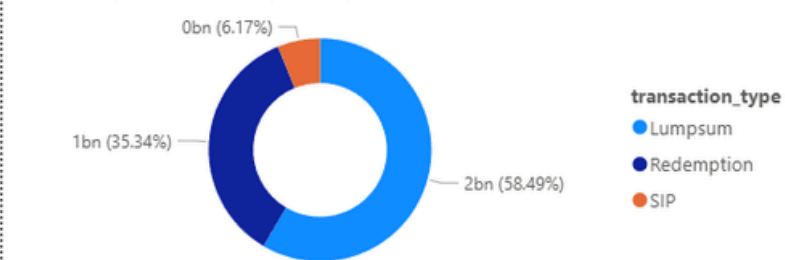
◆ AUM Trend Analysis

◆ Fund Scorecard

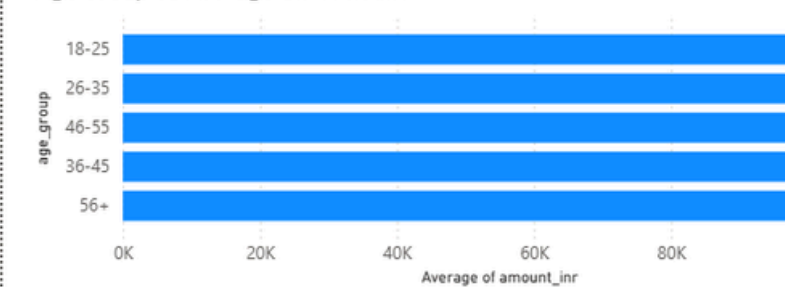
Dashboard Overview – Investor & Market Analytics

Power BI Dashboard – Page 3 & 4

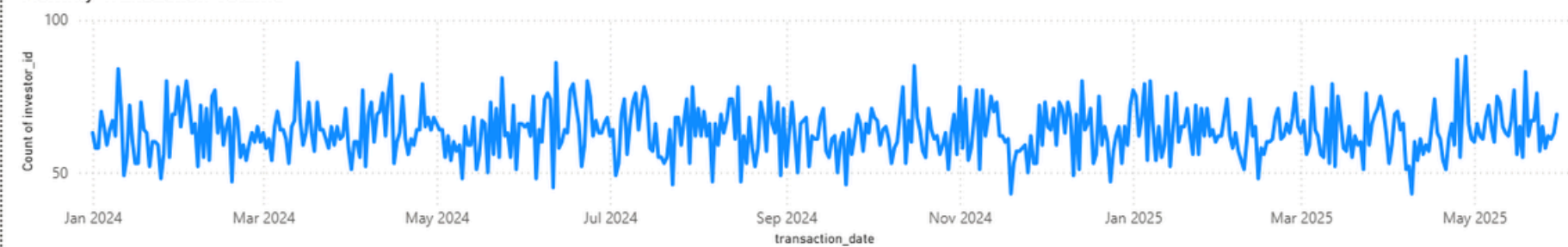
SIP / Lumpsum / Redemption Split



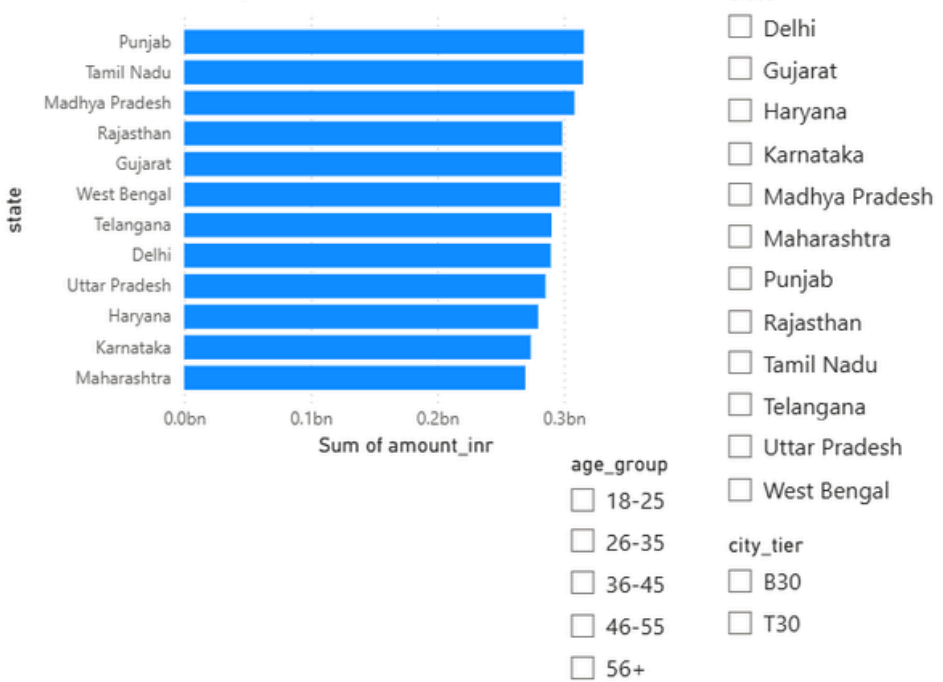
Age Group vs Average SIP Amount



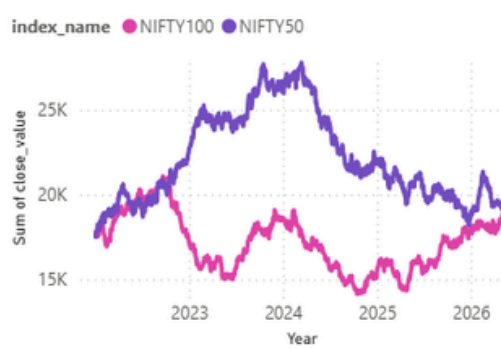
Monthly Transaction Volume



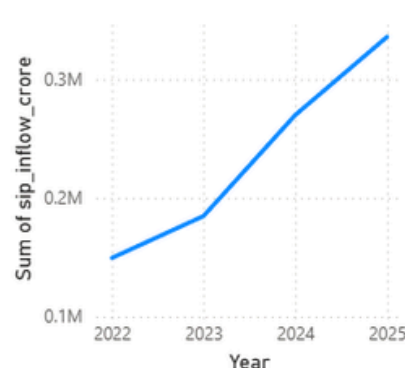
Transaction Amount by State



NIFTY Trend



SIP TREND

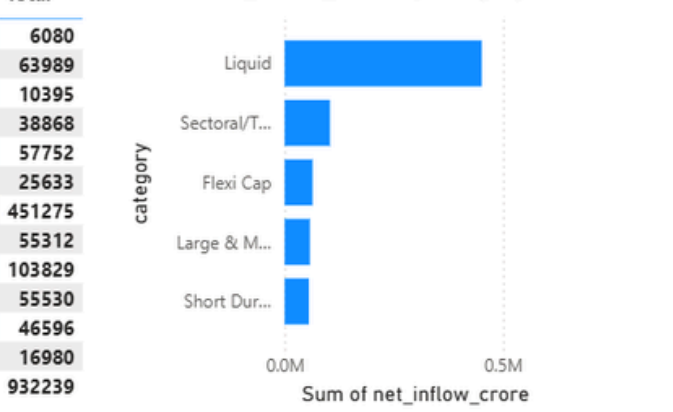


category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	41116	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Small Cap	34204	12392	46596
Value/Contra	12744	4236	16980
Total	706618	225621	932239

Highest Monthly SIP

31K
Highest Monthly SIP

Sum of net_inflow_crore by category



- category
- ☐ ELSS
 - ☐ Flexi Cap
 - ☐ Gilt
 - ☐ Hybrid
 - ☐ Large & Mid Cap
 - ☐ Large Cap
 - ☐ Liquid
 - ☐ Mid Cap
 - ☐ Sectoral/Thematic
 - ☐ Short Duration
 - ☐ Small Cap
 - ☐ Value/Contra

month



Business Value

Helps understand investor behavior patterns

Supports strategic decision-making

Dashboard Features

◆ State-wise Investment Analysis

◆ SIP vs Lumpsum Distribution

◆ Investor Demographics

◆ Market Trend Monitoring

Key Findings & Recommendations

Key Findings

1. SIP inflows reached record highs by 2025
2. Equity-oriented funds generated superior long-term returns
3. Higher Sharpe ratio funds delivered better risk-adjusted performance
4. Most investors maintained healthy SIP continuity
5. Portfolio concentration varied significantly across funds

Recommendations

- Focus on funds with strong Sharpe ratios and positive Alpha
- Monitor concentrated portfolios for sector-specific risks
- Improve investor engagement in underrepresented regions
- Promote SIP investing for long-term wealth creation

Conclusion

Successfully developed an end-to-end Mutual Fund Analytics platform covering:

Data Engineering

Financial Analytics

Risk Analytics

Business Intelligence

Dashboard Reporting

The solution provides actionable insights for investors, analysts, and fund managers.

Thank You